TUTORIAL XII

CROSS REFERENCES IN LaTeX

XII.1. WHY CROSS REFERENCES?

Cross reference is the technical term for quoting yourself. This is what you do when you say something like, "As I said earlier,...". More seriously, in a written article you may often have occasion to refer the reader to something mentioned earlier (or sometimes to something yet to be said) in the same document. Thus you may have explained a new term in the second section of your article and when you use this term again in the fourth section, it is a matter of courtesy to the reader to point to the explanation. Again, in a mathematics article, you may have to cite an earlier result in the proof of the current result.

Such cross referencing can be done by hand, but if you revise your document and insert some new sections (or theorems) then changing all cross references manually is no easy task. It is always better to automate such tedious tasks. (After all what's a computer for, if not to do such mundane jobs?)

XII.2. LET LaTeX DO IT

The basic method of using cross references (see Section XII.1 for what we mean by cross reference) in LaTeX is quite simple. Suppose that somewhere in the second section of your article, you want to refer to the first section. You assign a *key* to the first section using the command

```
\section{section name}\label{key}
```

and at the point in the second section where the reference is to be made, you type the command

```
\ref{key}
```

Thus the reference "see Section XII.1..." in the first sentence of this section was produced by including the command `\label{intro}` in the command for the first section as

```
\section{Why cross references}\label{intro}
```

and the command `\ref{intro}` at the place of reference in the second section as

```
...(see Section \ref{intro} for...
```

Okay, the example is a bit silly, since the actual reference here is not *really* necessary, but you get the general idea, don't you? Incidentally, the `\label{key}` for a section need not be given immediately after the `\section{section name}`. It can be given anywhere within the section.

The first time you run LaTeX on a file named, say, `myfile.tex` containing cross references, the reference information is written in an auxiliary file named `myfile.aux` and at the end of the run LaTeX prints a warning

```
LaTeX Warning: There were undefined references.

LaTeX Warning: Label(s) may have changed.
            Rerun to get cross-references right.
```

A second run gets the references right. The same thing happens when you've changed the reference information in any way, say, by adding a new section.

Though the *key* in `\label{`*key*`}` can be any sequence of letters, digits or punctuation characters, it is convenient to use some mnemonic (such as `\label{`*limcon*`}` for a section entitled "Limits and Continuity" rather than `\label{`*sec@#*?!*`}`. Also, when you make a reference, it's better to type `~\ref{`*limcon*`}` (notice the *tie*?) than `\ref{`*limcon*`}` to prevent the possibility of the reference number falling off the edge as in "...see Section XII.1 for further details....".

In addition to sectioning commands such as `\chapter` or `\section`, reference can also be made to an `\item` entry in an `enumerate` environment, by attaching a `\label`. For example the input

```
In the classical \emph{syllogism}
 \begin{enumerate}
 \item All men are mortal.\label{pre1}
 \item Socrates is a man.\label{pre2}
 \item So Socrates is a mortal.\label{con}
 \end{enumerate}
Statements (\ref{pre1}) and (\ref{pre2}) are the \emph{premises} and
statement (\ref{con}) is the conclusion.
```

gives the following output

In the classical *syllogism*

(1) All men are mortal.
(2) Socrates is a man.
(3) So Socrates is a mortal.

Statements (1) and (2) are the *premises* and statement (3) is the conclusion

You must be a bit careful about references to tables or figures (technically, "floats"). For them, the `\label` command should be given after the `\caption` command or in its argument, as in the example below:

```
\begin{table}[h]
\begin{center}
\setlength{\extrarowheight}{5pt}
\begin{tabular}{|c|c|c|c|}
\hline
Value of $x$ & 1 & 2 & 3\\
\hline
Value of $y$ & 1 & 8 & 27\\
\hline
\end{tabular}
\caption{Observed values of $x$ and $y$}\label{tabxy}
```

```
\end{center}
\end{table}
Two possible relations betweeen $x$ and $y$ satisfying
the data in Table\ref{tabxy} are $y=x^3$ and
$y=6x^2-11x+6$
```

This produces the following output:

Value of x	1	2	3
Value of y	1	8	27

Table XII.1: Observed values of x and y

Two possible relations between x and y satisfying the data in Table XII.1 are $y = x^3$ and $y = 6x^2 - 11x + 6$

You can think of a `\caption` command within a `figure` or `table` environment as a sort of sectioning command within the environment. Thus you can have several `\caption` and `\label` pairs within a single `figure` or `table` environment.

You can also make *forward* references in exactly the same way by `\ref`-ing to the *key* of some succeeding `\label` such as "see Subsection XII.2.1 for a discussion of cross references in mathematics."

XII.2.1. Cross references in math

Mathematical documents abound in cross references. There are references to theorems and equations and figures and whatnot. The method of reference is exactly as before. Thus if you've defined `\newtheorem{theorem}[subsection]`, then after typing

```
\begin{theorem}\label{diffcon}
Every differentiable function is continuous
\end{theorem}
```

you get

XII.2.1.1 Theorem. *Every differentiable function is continuous*

and you can type elsewhere in the document

```
The converse of Theorem~\ref{diffcon} is false.
```

to get

The converse of Theorem XII.2.1.1 is false.

References can be made to equations as in the following examples:

```
\begin{equation}\label{sumsq}
  (x+y)^2=x^2+2xy+y^2
\end{equation}
Changing $y$ to $-y$ in Equation~(\ref{sumsq}) gives the following
```